

STEFANSSON ISLAND, NU, Canada

WMO: 710170

Lat: 73.766N

Lon: 105.296W

Elev: 36

StdP: 14.68

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.7	-41.1	-51.6	0.3	-43.5	-48.9	0.3	-41.1	31.5	-11.6	29.2	-15.8	8.3	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	54.0	49.6	48.3	45.2	43.4	41.2	49.9	53.9	45.4	48.3	41.3	43.3	14.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
46.4	46.7	54.0	42.7	40.4	48.2	39.1	35.2	42.9	20.3	54.0	17.9	48.3	15.9	43.5	61.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.2	25.3	22.1	DB	-46.2	62.5	2.6	3.8	-48.1	65.2	-49.6	67.4	-51.1	69.5	-52.9	72.3	(4)
(5)			WB	-46.2	55.2	2.2	2.2	-47.7	56.8	-49.0	58.1	-50.3	59.3	-51.9	60.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-22.6	-25.1	-23.6	-9.2	14.5	32.3	37.7	35.3	25.8	10.4	-6.6	-19.0	(6)	
	DBStd	25.07	9.92	8.87	10.88	8.58	9.68	4.79	4.92	5.33	6.00	9.81	9.95	10.23	(7)	
	HDD50	16677	2250	2103	2283	1775	1102	530	386	458	725	1227	1698	2140	(8)	
	HDD65	22146	2715	2523	2748	2225	1567	980	848	920	1175	1692	2148	2605	(9)	
	CDD50	6	0	0	0	0	0	0	3	3	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	10.3	10.8	10.8	11.7	10.6	9.8	9.5	9.5	9.9	11.2	10.3	9.4	9.8	(14)	
	PrecAvg	5.7	0.2	0.1	0.2	0.2	0.4	0.6	1.0	1.1	0.8	0.5	0.3	0.2	(15)	
(16) Precipitation	PrecMax	9.2	0.4	0.5	0.5	0.7	1.2	1.7	2.0	3.1	2.1	1.3	0.6	0.5	(16)	
	PrecMin	2.1	0.0	0.0	0.1	0.0	0.0	0.1	0.2	0.3	0.2	0.1	0.0	0.0	(17)	
	PrecStd	1.3	0.1	0.1	0.1	0.2	0.2	0.4	0.4	0.7	0.4	0.3	0.2	0.1	(18)	
	PrecStd	1.3	0.1	0.1	0.1	0.2	0.2	0.4	0.4	0.7	0.4	0.3	0.2	0.1	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	4.9	12.1	18.6	35.3	52.0	62.0	61.1	42.1	31.0	22.1	19.8	(19)	
		MCWB	6.2	4.4	11.6	18.1	32.7	47.6	53.6	54.4	40.1	30.2	21.6	19.6	(20)	
	2%	DB	1.0	-2.1	6.1	12.6	32.6	45.7	55.8	53.6	38.1	28.7	16.2	5.9	(21)	
		MCWB	0.2	-2.6	5.6	11.6	30.9	43.0	51.0	49.7	36.8	28.1	15.6	5.3	(22)	
	5%	DB	-3.1	-5.9	-0.7	8.3	30.7	41.0	50.2	47.4	34.8	26.4	11.9	1.6	(23)	
		MCWB	-3.6	-6.1	-1.1	7.5	29.3	39.0	46.8	44.5	33.9	25.7	11.3	1.1	(24)	
	10%	DB	-7.9	-9.9	-6.8	4.4	28.1	37.7	44.7	42.3	32.7	23.9	7.3	-3.7	(25)	
		MCWB	-8.3	-10.3	-7.2	3.6	26.8	35.9	42.0	40.5	31.7	23.1	6.7	-4.2	(26)	
	0.4%	WB	6.0	4.5	11.4	18.2	33.0	47.7	53.9	55.2	40.4	30.5	21.6	19.3	(27)	
		MCDB	6.6	4.9	11.9	18.6	34.6	51.7	60.5	61.1	41.8	30.8	22.0	19.8	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	0.6	-2.7	5.5	11.7	31.3	42.9	51.0	49.6	36.9	28.1	15.8	5.5	(29)	
		MCDB	1.1	-2.4	5.9	12.5	32.5	45.7	55.9	53.5	38.0	28.6	16.2	5.9	(30)	
	5%	WB	-3.6	-6.2	-1.1	7.7	29.5	39.1	46.9	44.6	34.0	25.7	11.3	1.1	(31)	
		MCDB	-3.1	-6.0	-0.7	8.3	30.7	40.8	50.0	47.2	34.6	26.3	11.7	1.5	(32)	
	10%	WB	-8.3	-10.3	-7.1	3.7	26.8	36.0	42.3	40.5	31.9	23.2	6.8	-4.1	(33)	
		MCDB	-8.0	-10.0	-6.7	4.4	28.1	37.4	44.4	42.3	32.6	23.8	7.2	-3.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.3	13.0	14.8	11.6	7.0	8.2	7.6	5.8	10.0	12.0	12.0	(35)	
		MCDBR	15.3	17.0	17.0	13.8	11.3	12.5	18.4	17.1	8.6	9.6	14.6	15.0	(36)	
		MCWBR	15.0	16.7	16.8	13.3	10.1	10.4	14.7	14.0	7.7	9.5	14.7	14.9	(37)	
	5% WB	MCDBR	15.5	16.8	16.7	13.6	10.7	12.2	18.2	16.6	8.4	9.5	14.7	14.9	(38)	
		MCWBR	15.2	16.6	16.5	13.1	9.7	10.2	14.7	13.9	7.7	9.4	14.7	14.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.162	0.232	0.284	0.309	0.346	0.328	0.319	0.270	0.181	N/A	N/A	(40)	
	taud		N/A	1.842	2.107	2.048	2.056	2.085	2.309	2.366	2.345	1.876	N/A	N/A	(41)	
	Ebn at Noon		0	118	227	255	266	258	261	246	217	94	0	0	(42)	
	Edh at Noon		0	8	20	33	39	40	30	25	17	7	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0	40	431	1278	1903	2092	1688	967	423	99	0	0	(44)	
	RadStd		0	3	24	43	115	150	126	81	45	10	0	0	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page